

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Experiments

Course Code:	ITA05	Course Title:	Computer Vision
CO Assessed:	CO3 – Precision (accurate feature extraction & analysis) (BL3)	Assessment:	Assessment Tool 1 – Experiments
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	<i>Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BTL 4)</i>		
Student Name:	Kaviya V	Reg. No.:	192421022
Section / Batch:	Third Year/ B.Tech IT	Date:	11/08/2026

Code of Conduct

I KAVIYA V(192421022) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Kaviya V

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts	
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation	
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools	
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results	
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation	

CO3 - Assessment Tool 1 - Experiments

6. Hough Transform for Shape Detection

Scenario: Conduct an experiment using the Hough Transform to detect lines or circles in an image. Explain the principles and significance of the Hough Transform in shape detection. Execute the detection process using appropriate tools, record the detected shapes and their parameters, and analyze the detection accuracy under different image conditions.

CODE:

```
import cv2
import numpy as np

img = cv2.imread(
    r"C:\Users\Kaviya V\OneDrive\Documents\A~COLLEGE\CV\EXPS\image.jpg")

if img is None:
    print("Error: Image not found!")
    exit()

original = img.copy()
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
edges = cv2.Canny(gray, 50, 150)

# Hough Line Transform
lines = cv2.HoughLinesP(
    edges,
    1,
    np.pi / 180,
    threshold=50,
    minLineLength=50,
    maxLineGap=10
)

# Draw detected lines
if lines is not None:

    # Convert lines into (N, 4) format
    lines = np.array(lines).reshape(-1, 4)

    print("Detected Lines:")

    for line in lines:
        x1, y1, x2, y2 = map(int, line)
```

```
print(x1, y1, x2, y2)

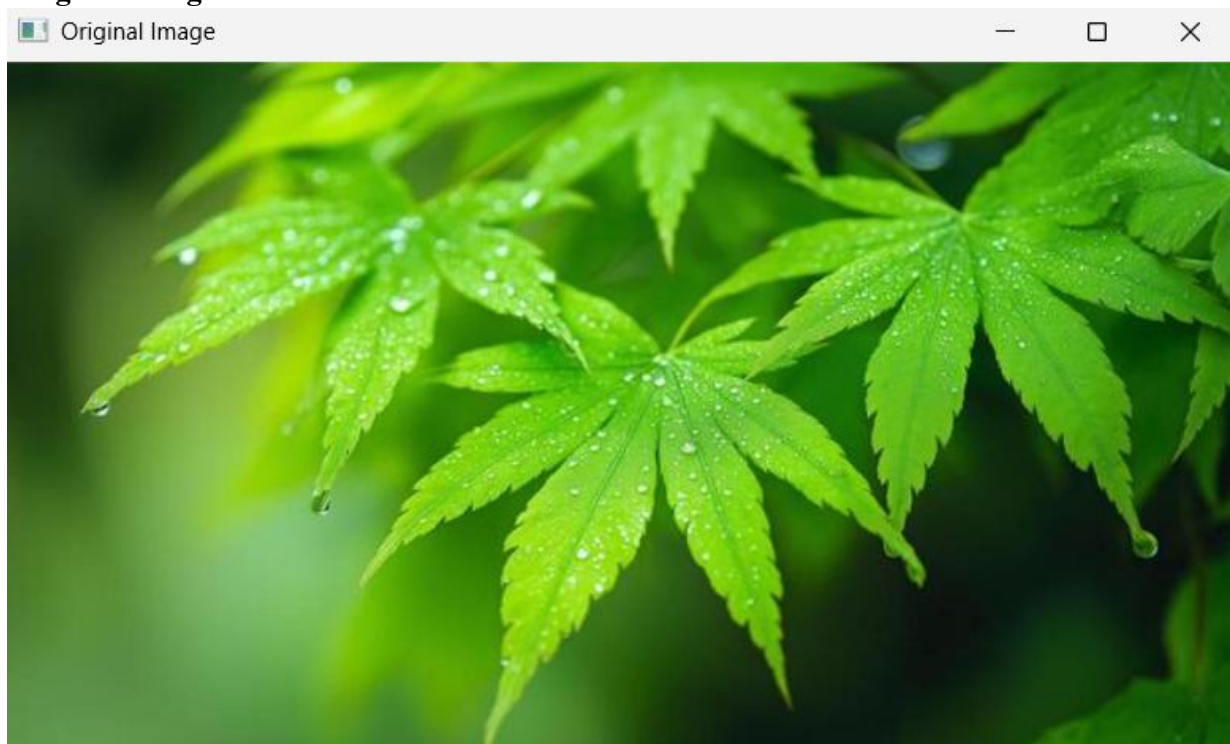
cv2.line(
    img,
    (x1, y1),
    (x2, y2),
    (0, 0, 255),
    2
)

else:
    print("No lines detected.")

# Display results
cv2.imshow("Original Image", original)
cv2.imshow("Edges", edges)
cv2.imshow("Hough Lines", img)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

Original Image:



OUTPUT:

```
*IDLE Shell 3.14.3*
File Edit Shell Debug Options Window Help
>>> ===== RESTART: C:/Users/Kaviya V/OneDrive/Documents/A~COLLEGE/CV/6 ass.py =====
Detected Lines:
408 107 512 75
430 125 489 66
77 133 136 129
418 126 561 66
441 110 529 107
186 251 265 212
320 177 569 76
297 177 376 140
424 122 529 73
306 147 396 208
245 201 368 141
314 163 364 142
214 114 281 115
126 115 189 112
299 175 370 141
544 104 613 83
449 130 568 84
220 200 272 176
334 178 407 141
450 128 509 124
297 180 378 146
298 181 382 147
497 109 607 131
250 175 372 137
246 204 381 149
524 69 582 51
275 302 320 247
99 108 196 100
427 107 493 97
292 156 400 213
422 119 498 79
221 214 380 150
468 86 553 62
437 99 506 82
416 130 489 100
406 100 484 74
260 125 399 212
```

Edges:



Hough Lines:

